Pure Mathematics with Mechanics GCE A Level 2024 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths with Mechanics 1, June 2024, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Formulae booklets and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room. The scan is complete: cover page and five pages of questions (pages 2 to 6), questions numbered 1 to 50.

SECTION A: PURE MATHEMATICS

Given the functions $f, g : \mathbb{R} \to \mathbb{R}$ where $f(x) = x^2 - 1$ and $g(x) = 2x + 1$, $(g \circ f)(2) =$

a) 11
b) 7
c) 24
d) 13

2. If $\log_x y = 2$ and $xy = 125$, then the values of x and y are respectively

a) 3 and 9
b) 9 and 3
c) 5 and 25
d) 25 and 5

3. The vertical asymptotes of the curve $y = \dfrac{1}{(x-2)(x+3)}$ are

a) $x = 3,\ x = -2$
b) $x = 2,\ x = -3$
c) $x = -3,\ x = -2$
d) $x = 2,\ x = 3$

4. When the polynomial $p(x) = ax^3 + 5x^2 + 2x - 8$ is divided by $(x - 1)$, the remainder is -3. $a =$

a) -4
b) -2
c) 2
d) 3

5. The set of values of x for which $\dfrac{(x+1)^2 + 3}{(x+3)(x-2)} < 0$ is

a) $\{x : x < -3 \text{ or } x > 2\}$
b) $\{x : x \leq -3 \text{ or } x > 2\}$
c) $\{x : -3 \leq x < 2\}$
d) $\{x: -3

6. A function $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 4\beta + 4x, & x < 1 \\ 4x - \beta^2 - 4, & x > 1 \end{cases}$$

If $f(x)$ has a limit at $x = 1$, the value of β is

a) 2
b) 1
c) 3
d) -2

7. Let p and q be statements.

p	q	$\oplus$
T	T	T
T	F	T
F	T	T
F	F	F

The compound statement $\oplus$ on the table could represent

a) $p \to q$
b) $p \leftrightarrow q$
c) $p \vee q$
d) $p \wedge q$

8. Given the lines

$L_1 : \ \mathbf{r} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k} + \lambda(4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$

$L_2 : \ \mathbf{r} = \mathbf{i} + 3\mathbf{j} + \mathbf{k} + \delta(8\mathbf{i} + 4\mathbf{j} - 6\mathbf{k})$

$L_3 : \ \mathbf{r} = 2\mathbf{i} + 6\mathbf{j} + 2\mathbf{k} + \beta(\mathbf{i} - 4\mathbf{j} + 4\mathbf{k})$

$L_4 : \ \mathbf{r} = 4\mathbf{i} + 12\mathbf{j} + 4\mathbf{k} - \mu(\mathbf{i} + 3\mathbf{j} + \mathbf{k}).$

Which pair of lines are parallel?

a) L_1 and L_2
b) L_1 and L_4
c) L_2 and L_4
d) L_2 and L_3

9. A relation R defined on a non-empty set S is called a partial order if R is

a) Reflexive, symmetric and anti-symmetric
b) Reflexive, anti-symmetric and transitive
c) Reflexive, symmetric and transitive
d) Symmetric, anti-symmetric and transitive

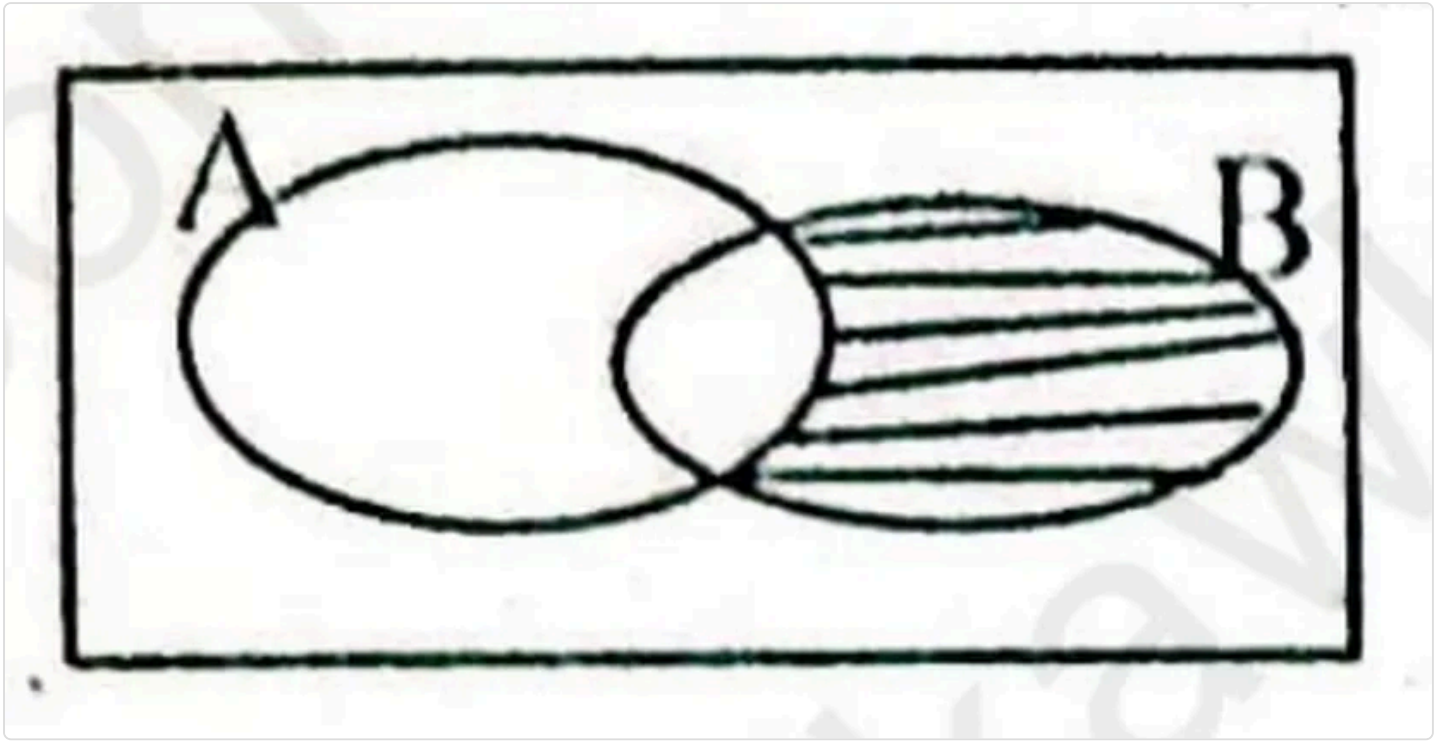
10. If $\sum_{r=1}^{\infty} \frac{m}{2^{r-1}} = 10$, then $m =$

a) 10
b) 5
c) 15
d) 7

11. The first order differential equation obtained from the function $y = Ax + \ln x$ is

a) $x\frac{dy}{dx} = y + \ln x + 1$
b) $\frac{dy}{dx} = y - \ln x + 1$
c) $x\frac{dy}{dx} = y - \ln(x + 1)$
d) $x\frac{dy}{dx} = y - \ln x + 1$

12. The shaded region represents



a) $A \cap B$
b) $A \cap B'$
c) $A' \cap B$
d) $(A \cup B)'$

13. A function $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = \frac{x + 6}{x^2 - 2x - 15}$. The domain of f is

a) $\mathbb{R} - \{3, 5\}$
b) $\mathbb{R} - \{-3, 5\}$
c) $\mathbb{R} - \{-5, 3\}$
d) $\mathbb{R} - \{-5, -3\}$

14. A curve is defined parametrically by $x = 2t,\ y = \dfrac{2}{t}$. The Cartesian equation of the curve is

a) $xy = 4$
b) $xy = 2$
c) $xy = 1$
d) $xy = 3$

15. If $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{s} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, then $\mathbf{r} \times \mathbf{s}$ is

a) $-4\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}$
b) $-4\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$
c) $-4\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$
d) $-4\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$

16. Given that $f(\theta) = \sin\theta - \sqrt{3}\cos\theta$. Expressing $f(\theta)$ in the form $R\sin(\theta - \alpha)$ where $R > 0$ and α is an acute angle, we obtain $f(\theta) =$

a) $2\sin\left(\theta - \dfrac{\pi}{2}\right)$
b) $\sqrt{10}\sin\left(\theta - \dfrac{\pi}{3}\right)$
c) $2\sin\left(\theta - \dfrac{\pi}{3}\right)$
d) $\sqrt{10}\sin\left(\theta - \dfrac{\pi}{4}\right)$

17. If $f(x) = x^2 + kx - 3$ has a root in the interval $[1, 2]$, then the range of k is

Options as printed: A) $k < -\dfrac{1}{2}$ B) $k > 2$ C) $-\dfrac{1}{2}$

18. Given that the sum and product of roots of the equation $2x^2 - 2x + q = 0$ are equal, the value of q is

a) -2
b) $\dfrac{1}{2}$
c) 2
d) $-\dfrac{1}{2}$

19. The sum of the first n terms of an arithmetic progression is $n(2n + 3)$. The nth term of the progression is

a) $6n + 1$
b) $2n + 1$
c) $4n - 1$
d) $4n + 1$

20. $\left|\dfrac{-5 + 10i}{1 + 2i}\right| =$

a) 5
b) $\sqrt{5}$
c) $\sqrt{125}$
d) 6

21. The line L has equation $\mathbf{r} = 5\mathbf{i} - 8\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ and the plane π has equation $2x + 2y - z = 5$. The sine of the acute angle between the line and the plane is

a) $\dfrac{3\sqrt{14}}{7}$
b) $\dfrac{\sqrt{14}}{7}$
c) $\dfrac{2\sqrt{14}}{7}$
d) $\dfrac{4\sqrt{14}}{7}$

22. $\dfrac{d}{dx}(3x + 2)^4 =$

a) $12x(3x + 2)^3$
b) $12(3x + 2)^4$
c) $12(3x + 2)^3$
d) $\dfrac{4}{3}(3x + 2)^3$

23. The equation of the circle which passes through the origin and the points $(2, 0)$ and $(0, -4)$ is

a) $x^2 + y^2 - 2x + 4y = 0$
b) $x^2 + y^2 - 2x - 4y = 0$
c) $x^2 + y^2 + 2x - 4y = 0$
d) $x^2 + y^2 + 2x + 4y = 0$

24. The area of the finite region bounded by the curve $y = 2x - \dfrac{1}{x}$ and the lines $x = 1$ and $x = 3$ is

a) $8 - \ln 3$
b) $8 + \ln 3$
c) $8 - \ln 2$
d) $8 + \ln 2$

25. If a root of the equation $x^2 + 2x + 1 = 0$ is also a root of the equation $x^3 - 2x^2 + 6x + k = 0$, the value of the constant k is

a) 9
b) -9
c) 5
d) -5

26. A periodic function f of period 4 is defined by

$$f(x) = \begin{cases} x^2, & 0 \le x < 2 \\ x + 2, & 2 \le x < 4 \end{cases}$$

$f(9) =$

a) 1
b) 3
c) 11
d) 81

27. Two functions $f, g : \mathbb{R} \to \mathbb{R}$ are such that $f(x) = 2x - 1$ and $(f \circ g)(x) = x - 2$. Then $g(x) =$

a) $\dfrac{x-2}{2}$
b) $\dfrac{x-1}{2}$
c) $\dfrac{2x-1}{2}$
d) $\dfrac{x+1}{2}$

28. The equation of the tangent to the curve $y = 3x(2 - x^2)$ at the point $(1, 3)$ is

a) $3x - y - 6 = 0$
b) $3x + y - 6 = 0$
c) $y - 3x - 6 = 0$
d) $3x - y + 6 = 0$

29. Simplifying $\dfrac{\sin 5x - \sin x}{\sin 4x - \sin 2x}$ gives

a) $2\cos x$
b) $2\cos 2x$
c) $2\sec x$
d) $2\sin x$

30. The number of arrangements of the letters of the word $ARREARS$ is

a) 210
b) 420
c) 640
d) 840

31. In the expansion of $\sqrt{3 + x}$, the range of values of x for which the complete expansion is valid is

a) $-3 \leq x \leq 3$
b) $-3 c) $-1 d) $-\dfrac13$

32. Evaluating $1 + \dfrac{1}{3} + \dfrac{1}{9} + \dfrac{1}{27} + \cdots$ gives

a) $\dfrac{1}{3}$
b) $\dfrac{3}{2}$
c) $\dfrac{40}{27}$
d) 3

33. $\lim\limits_{x \to \infty} \left(\dfrac{4x^2 - 1}{2x^2 + 1} \right)$ is

a) 0
b) 1
c) 2
d) 3

34. $\int_1^4 \left(\dfrac{x + 3}{x} \right) dx =$

a) $3 + 4 \ln 3$
b) $3 + 3 \ln 4$
c) $4 + 4 \ln 3$
d) $4 + 3 \ln 4$

35. If the matrix $\mathbf{M} = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -1 & 2 \\ -1 & 3 & 4 \end{pmatrix}$, then the determinant of $\mathbf{M}$ is

a) -18
b) -8
c) 0
d) 18

SECTION B: MECHANICS

The position vector of a particle at time t seconds is $\mathbf{r}$, where $\mathbf{r} = \left[(t^2 + 1)\mathbf{i} + t^3\mathbf{j} \right]$ m. The velocity of the particle when $t = 2$ is

a) $(4\mathbf{i} + 6\mathbf{j})$ m s^{-1}
b) $(4\mathbf{i} - 12\mathbf{j})$ m s^{-1}
c) $(2\mathbf{i} + 12\mathbf{j})$ m s^{-1}
d) $(4\mathbf{i} + 12\mathbf{j})$ m s^{-1}

37. The extension a force of 20 N produces in a spring of natural length 5 m whose modulus of elasticity is 150 N is

a) $\dfrac{3}{2}$ m
b) $\dfrac{3}{4}$ m
c) $\dfrac{2}{3}$ m
d) $\dfrac{1}{2}$ m

38. The velocity of a particle A relative to another B is $(7\mathbf{i} + 5\mathbf{j})$ m s^{-1}. Given that the velocity of A is $(8\mathbf{i} + 6\mathbf{j})$ m s^{-1}. The velocity of B is

a) $(15\mathbf{i} + 11\mathbf{j})$ m s^{-1}
b) $(-\mathbf{i} - \mathbf{j})$ m s^{-1}
c) $(\mathbf{i} + \mathbf{j})$ m s^{-1}
d) $(15\mathbf{i} - 11\mathbf{j})$ m s^{-1}

39. Given that A and B are events with $P(A) = \frac{1}{4}, P(A \cup B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{12}. P(A \mid B) =$

a) $\frac{1}{2}$
b) $\frac{1}{3}$
c) $\frac{1}{6}$
d) $\frac{2}{3}$

40. A small sphere of mass $0.6\ \text{kg}$ is moving on a horizontal table with a speed of $10\ \text{m s}^{-1}$ when it collides with a fixed smooth wall perpendicular to the table surface. It rebounds with a speed of $\frac{15}{2}\ \text{m s}^{-1}$. The coefficient of restitution between the sphere and the wall is

a) $\frac{4}{3}$
b) $\frac{5}{3}$
c) $\frac{4}{5}$
d) $\frac{3}{4}$

41. Three particles of masses $5\ \text{kg}$, $3\ \text{kg}$ and $m\ \text{kg}$ lie on the y-axis at the points $(0, 4)$, $(0, 2)$ and $(0, 5)$ respectively. The center of mass of the system is at the point $(0, 4)$. The value of m is

a) 5
b) 4
c) 6
d) 7

42. A car wheel of radius $\frac{1}{2}\ \text{m}$ is moving at a speed of $\frac{10}{3}\ \text{m s}^{-1}$. Its angular speed in rad s^{-1} is

a) $\frac{20}{3}$
b) $\frac{5}{3}$
c) $\frac{16}{3}$
d) $\frac{25}{3}$

43. A car accelerates uniformly from a velocity of $10\ \text{m s}^{-1}$ to $60\ \text{m s}^{-1}$ in 6 seconds on a straight road. The distance it covered is

a) $150\ \text{m}$
b) $200\ \text{m}$
c) $210\ \text{m}$
d) $215\ \text{m}$

44. A box contains ten green and six white marbles. A marble is chosen at random, its colour noted and it is not replaced. This process is repeated once. The probability that the marbles chosen are of the same colour is

a) $\frac{17}{32}$
b) $\frac{1}{3}$
c) $\frac{1}{2}$
d) $\frac{13}{32}$

45. The engine of a car of mass $1200\ \text{kg}$ works at a constant rate of $69\ \text{kW}$ up a road inclined at an angle $\sin^{-1}\left(\frac{1}{6}\right)$ to the horizontal. Given that the non-gravitational resistance to the motion of the car is $300\ \text{N}$ and taking g as $10\ \text{m}\,\text{s}^{-2}$, the maximum speed of the car is

a) $15\ \text{m}\,\text{s}^{-1}$
b) $25\ \text{m}\,\text{s}^{-1}$
c) $30\ \text{m}\,\text{s}^{-1}$
d) $22\ \text{m}\,\text{s}^{-1}$

46. The work done by a force $\mathbf{F} = (\mathbf{i} - 2\mathbf{j} - 4\mathbf{k})\ \text{N}$ which moves its point of application from a point A with position vector $\mathbf{r}_A = (3\mathbf{i} - 4\mathbf{j})\ \text{m}$ to another point B with position vector $\mathbf{r}_B = (-\mathbf{i} - 6\mathbf{k})\ \text{m}$ is

a) $47\ \text{J}$
b) $36\ \text{J}$
c) $12\ \text{J}$
d) $-6\ \text{J}$

47. A particle decelerating to rest has its speed $v\ \text{m}\,\text{s}^{-1}$ at time t seconds given by $v = 6t - \frac{1}{2}t^2$. The deceleration of the particle when $v = 10$ is

a) $4\ \text{m}\,\text{s}^{-2}$
b) $10\ \text{m}\,\text{s}^{-2}$
c) $-10\ \text{m}\,\text{s}^{-2}$
d) $-4\ \text{m}\,\text{s}^{-2}$

48. A uniform ladder AB, of length $2a$, rests in limiting equilibrium with its top end against a rough vertical wall and its lower end on a rough horizontal floor. The forces acting on the ladder are as shown in Fig. 1.

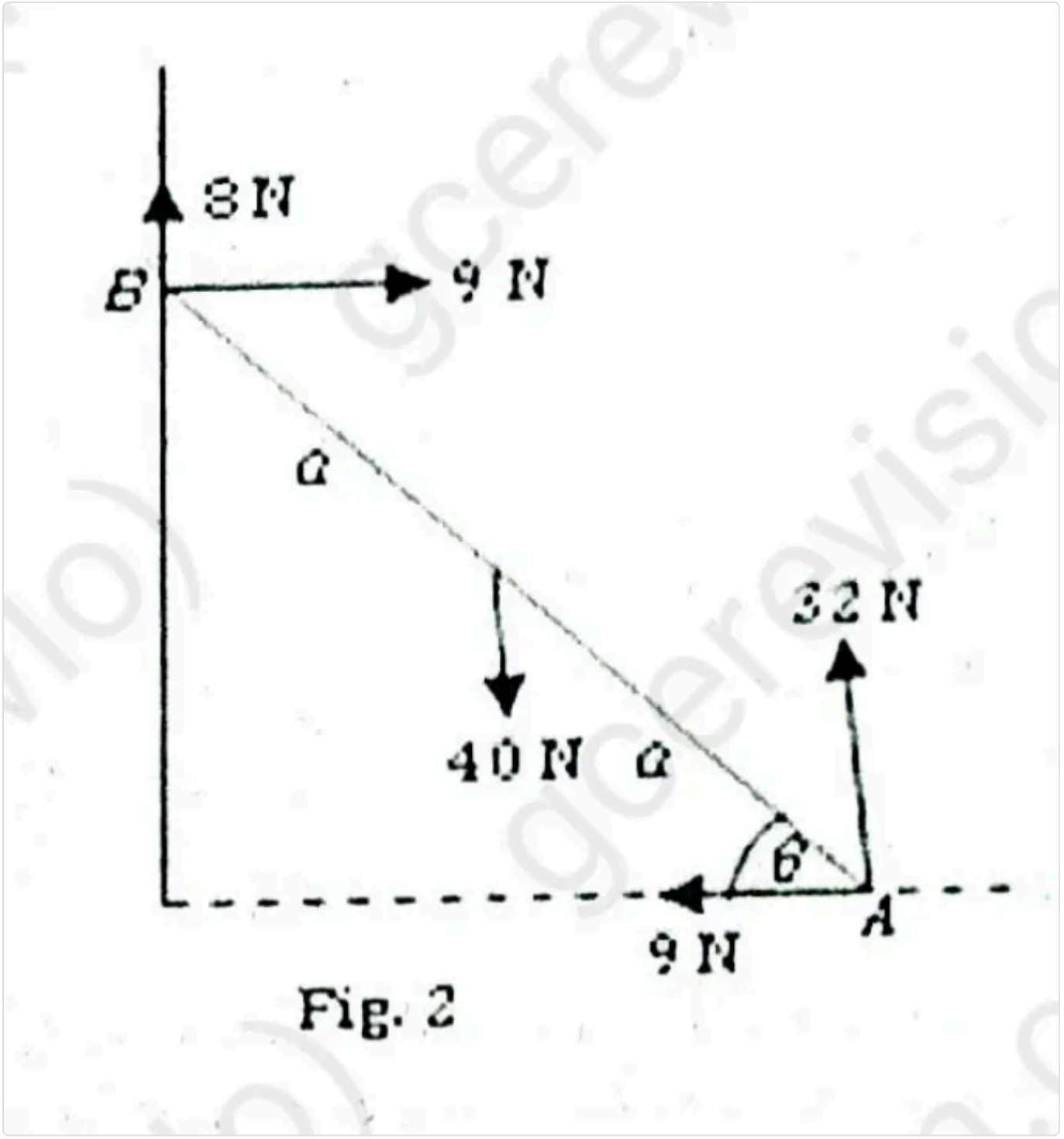


Fig. 2

The angle θ which AB makes with the floor is given by

a) $\tan\theta = \frac{4}{3}$

b) $\tan\theta = \frac{3}{4}$

c) $\sin\theta = \frac{4}{3}$

d) $\sin\theta = \frac{3}{4}$

49. A particle of mass $14\ \text{kg}$ rests on a smooth horizontal table. It is connected by a light inextensible string passing over a smooth pulley fixed at the edge of the table to a particle of mass $6\ \text{kg}$ which hangs freely. Given that the acceleration of the system is $3\ \text{m}\,\text{s}^{-2}$ when it is released from rest, the force exerted by the string on the pulley is

a) $42\sqrt{2}\ \mathrm{N}$
b) $82\sqrt{2}\ \mathrm{N}$
c) $42\ \mathrm{N}$
d) $82\ \mathrm{N}$

50. A smooth sphere A of mass $6\ \mathrm{kg}$ travelling at $4\ \mathrm{m\,s^{-1}}$ collides directly with a smooth sphere B of mass $3\ \mathrm{kg}$ travelling at $2\ \mathrm{m\,s^{-1}}$ in the opposite direction. Given that the speed of B after impact is $2\ \mathrm{m\,s^{-1}}$, the kinetic energy of A after impact is

a) $6\ \mathrm{J}$
b) $12\ \mathrm{J}$
c) $50\ \mathrm{J}$
d) $13.5\ \mathrm{J}$

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.